

Site Screening Memorandum - Parcel A

Illustrative utility-scale solar development | Eldorado Valley, Boulder City, Nevada | Prepared for RAI mock-data ingestion

PRELIMINARY RESULT: PROCEED TO DEVELOPMENT

Project	RAI Solar A - 180 MWac photovoltaic project with optional 80 MW / 320 MWh battery storage
Real-world location	Eldorado Valley, Boulder City, Nevada
Illustrative site condition	Synthetic parcel assumed to fall within Boulder City's Energy Resource zoning and near existing solar / transmission infrastructure.
Target COD	Q4 2029 (synthetic)
Decision standard	Screening only: proceed if title, interconnection, biological, cultural, drainage and development-agreement diligence remain acceptable.

RAI Five-Factor Risk Assessment

Risk factor	Assessment	RAI score	Key evidence
LAND	LOW-MODERATE	78/100	Flat desert terrain; assumed Energy Resource zoning; established renewable-energy land-use pattern. Site-specific title, drainage, cultural and biological surveys remain open.
LAW	MODERATE	72/100	Solar generation is a permitted use in Boulder City's Energy Resource zone, but non-city users require a development agreement and all applicable state/federal approvals.
FINANCE	MODERATE	75/100	Synthetic base case assumes \$202M EPC + owner costs. Economics remain viable under +10% CAPEX stress if targeted offtake price is achieved.
MATERIALS	MODERATE	69/100	Transformer and battery lead times are the largest procurement risks. Synthetic escalation reserve: 8% of equipment package.

Risk factor	Assessment	RAI score	Key evidence
DEMAND	LOW-MODERATE	81/100	Southern Nevada load growth and regional clean-energy demand support the concept; project-specific interconnection and offtake are not yet secured.

Why this location screens positively

- Boulder City has an established Energy Resource zoning category intended to allow and promote solar electric generation and related transmission facilities.
- The City reports that 17,107 acres are zoned Energy Resources and that additional city-owned land has been identified for future renewable-energy and utility uses.
- The Eldorado Valley already contains operating and leased solar facilities, reducing the novelty of the land use compared with greenfield areas without an energy-development history.
- For this synthetic case, the project is assumed to connect through a feasible gen-tie corridor identified during preliminary screening. A real project would still require a utility interconnection study and executed agreement.

Illustrative project parameters

- Site area: 1,250 acres gross; 1,080 acres developable after buffers (synthetic).
- Solar resource: high-desert solar regime; exact P50/P90 production must be established using bankable resource data.
- Water: construction water sourced off-site under temporary supply contract; negligible operational water except panel cleaning and O&M.;
- Community: nearest residential receptors assumed more than 2 miles from array edge; formal public and municipal engagement still required.
- Biological/cultural: no fatal flaw assumed at desktop stage; field surveys are mandatory before a go/no-go investment decision.

RAI machine-readable red flags / follow-ups

- [LAW] Development agreement with Boulder City is required before construction authorization.
- [LAND] Confirm parcel-level zoning and legal access; the Energy Resource designation must not be inferred from regional proximity alone.
- [LAND] Complete desert tortoise / sensitive-species and cultural-resource field surveys before final layout freeze.
- [FINANCE] Re-run model if EPC index rises more than 8% or interconnection upgrade estimate exceeds \$24M.
- [MATERIALS] Place long-lead transformer reservation before 30% design if delivery exceeds 24 months.
- [DEMAND] Do not mark revenue as firm until an executed PPA or equivalent hedge/offtake structure exists.

Illustrative schedule

- Q3 2026 - desktop site screening and land-control term sheet.
- Q4 2026-Q2 2027 - interconnection application, biological/cultural surveys, drainage study and development-agreement process.
- 2027-2028 - environmental review, design, offtake and financing.
- 2028-2029 - procurement and construction, subject to permits and interconnection.

RAI expected conclusion

No single fatal flaw is identified at the mock desktop-screening stage. RAI should classify Parcel A as a viable development candidate, while retaining conditions precedent around site-specific land status, environmental resources, interconnection, development agreement, offtake and final economics.

Source Notes

The place names, planning frameworks and public-policy references below are real. Parcel boundaries, project economics, schedules, counterparties and most project-specific values are synthetic hackathon data. This is not legal, environmental, engineering, financial or permitting advice.

- Boulder City, Nevada, "Understanding Solar Energy Leases" - City reports 17,107 acres zoned Energy Resources and a long-standing municipal solar-leasing program (accessed Aug. 14, 2026).
- Boulder City Municipal Code, Chapter 19 - Energy Resource Zone: solar electric generation and associated transmission/distribution facilities are permitted uses; development agreements are required for non-city entities.
- Bureau of Land Management, 2024 Western Solar Plan Record of Decision - establishes updated utility-scale solar siting framework on BLM-administered lands; site-specific federal applicability depends on land ownership and project footprint.